

ENVS-193DS Final

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[Link to repo](#)

```
# read in packages
library(tidyverse)
library(here)
library(DHARMA)
library(ggeffects)
library(rstatix)
library(janitor)

# reading in data and creating new object
nest <- read_csv(
  here("data", "nest_data_final.csv"))

# creating new data frame with personal data
personal_data <- read_csv(here("data", "personal-datasheet.csv")) |>
  # standardizing column names
  clean_names() |>
  # dropping rows with na observations
  drop_na() |>
  # changing the stop column to display TRUE/FALSE instead of 1/0
  mutate(stop = as.logical(stop),
         week = as.factor(week))
```

Problem 1. Research writing

a.

In part one they used a generalized linear model (specifically a logistic regression). The provided p-value indicates the predictor accurately explains the variance in the response.

In part two they used a Chi-square test to see if there was a relationship between the two categorical variables. The p-value indicates that there is a relationship between the two.

b.

Observed American Avocets (*Recurvirostra americana*) were more likely to utilize micro-habitats closer to the water's edge. This suggests that proximity to water is a key feature of habitat suitability for this species.

The three observed shorebird species (American Avocets, Forster's Tern, Black-necked Stilt) were not evenly distributed across habitats (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat). This means that each species tends to favor a specific habitat or multiple habitats, had there instead been even or completely random distribution we would not be able to assume this.

c.

- An additional variable that we could collect would be percent vegetation coverage in each micro-habitat.
- This would be measured as a continuous variable with values ranging from 0% to 100%
- This variable could influence American Avocet micro-habitat choice as more coverage could provide more protection from predators or could conversely reduce foraging efficiency for the birds due to the density of foliage. Data collection for this would be fairly easy by either using quadrats at each habitat site (and collected each plant species + coverage) or by taking pictures of each habitat and using software like ImageJ to see the extent of coverage.

Problem 2. Data Analysis

a.

```
# creating clean data frame
nests_clean <- nest |>
  # selecting desired columns
  select(ant_species, height_cm, case_control) |>
  # filtering for desired ant species
  filter(ant_species %in% c("RRB", "BBR", "AB")) |>
  # creating new column for full scientific name
  mutate(sci_name = case_when(
    ant_species == "AB" ~ "Crematogaster sjostedti",
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"),
    # reordering factors for ant species
    ant_species = factor(ant_species,
      levels = c("AB", "RRB", "BBR"))
  )

# displaying structure of data frame
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ ant_species : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ sci_name     : chr [1:270] "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimosae" ...
```

```
# slicing sample
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
  ant_species height_cm case_control sci_name
  <fct>         <dbl>         <dbl> <chr>
1 RRB           166.             0 Crematogaster mimosae
2 RRB           121             0 Crematogaster mimosae
3 RRB           216             1 Crematogaster mimosae
4 BBR            93             0 Crematogaster nigriceps
5 RRB           423             0 Crematogaster mimosae
6 RRB           210             0 Crematogaster mimosae
7 RRB           212.             1 Crematogaster mimosae
8 RRB           513             0 Crematogaster mimosae
9 RRB           174.             0 Crematogaster mimosae
10 RRB          300             0 Crematogaster mimosae
```

b.

For this data, a value of 1 in the case_control column indicates that a nest was observed in the acacia tree in question. A value of zero in that said column means that no bird nest was observed in the tree.

c.

Tree Height (height_cm)

Tree height in this data set is a continuous variable that is measured in centimeters. As tree height increases the probability of nest occurrence would likely increase. Taller trees are more structurally complex and therefore have more nesting opportunities, they may also provide better predator avoidance for birds (From Results paragraph 1 and Discussion paragraph 2).

Acacia ant species (ant_species)

Ant species is a categorical variable with four factors. As ant species changes, the probability of nest occurrence would likely differ. This is due to each ant species exhibiting a distinct aggression level and mutualistic relationship with the host Acacia tree which affects the level of predator protection a nest in said tree would receive. (Results, paragraphs 1 + 2, Introduction, paragraph 1, Discussion, paragraph 3)

d.

Model number	Ant Species	Tree Height	Predictor list
1	X	X	all predictors, no interaction (null model)
2	X	X	all predictors, with interaction (full model)

e.

f.

```
# running AIC test to determine best model
AIC(model1, model2)
```

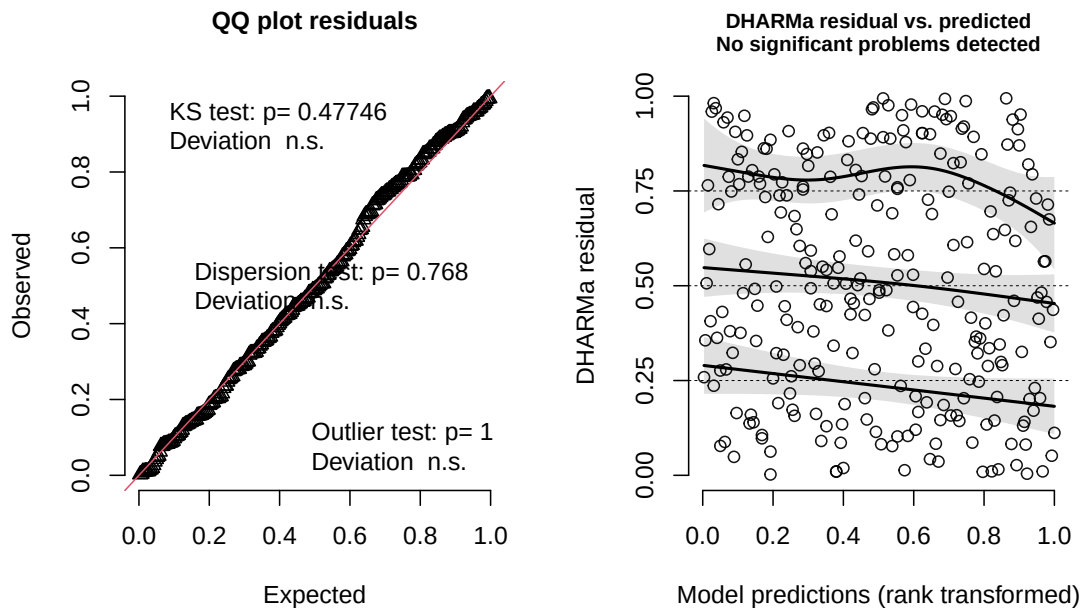
```
          df      AIC
model1  4 258.7430
model2  6 237.8042
```

The best model as determined by Akaike's Information Criterion (AIC) was the model that predicted test_case (whether or not a sample tree had a nest) using the interaction of tree height (height_cm) and species of acacia ant (ant_species)

g.

```
# plotting simulated residuals from DHARMA package.
plot(
  simulateResiduals(model2)
)
```

DHARMA residual



Simulated residuals look good, QQ has very little deviation from expected and residual vs predicted looks random with no discernible pattern.

h.

```
# creating prediction model for model2
preds <- ggpredict(model2,
  # terms for model, all tree heights all species
  terms = c("height_cm[all]", "ant_species")) |>
# changing grouping column's name
rename(ant_species = group) |>
arrange(x)

# setting custom species colors
sp_cols <- c(
  "AB" = "lightblue3",
  "RRB" = "tan",
  "BBR" = "orange"
)
```

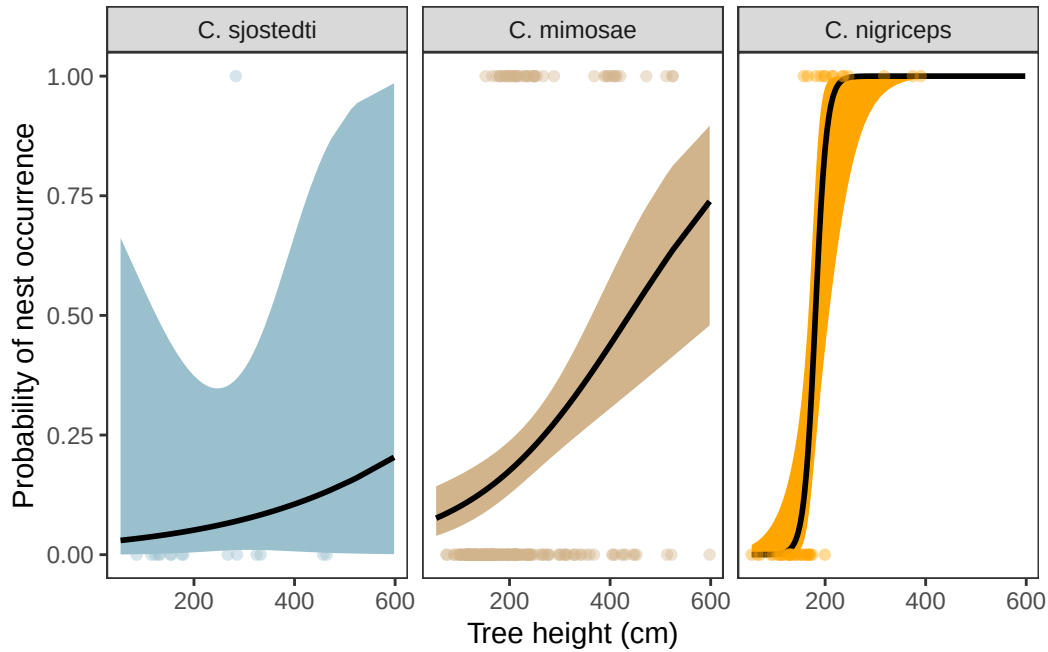
```

# base layer: ggplot
ggplot() +
# first layer: model confidence ribbon
geom_ribbon(data = preds, # prediction data frame
           aes(x = x,
               ymin = conf.low, # min by low confidence
               ymax = conf.high, # max by high confidence
               fill = ant_species)) + # filling color by group
# second layer: model probability line
geom_line(data = preds, # prediction data frame
          aes(x = x,
              y = predicted),
          # coloring line black
          color = "black",
          # making line thicker
          linewidth = 1) +
# third layer: jitter for underlying data
geom_jitter(data = nests_clean, # nests_clean data frame
            aes(x = height_cm, # x-axis: tree height
                y = case_control, # y-axis: nest occurrence
                color = ant_species), # coloring by species
            alpha = 0.4, # making points more opaque
            height = 0) + # removing vertical jitter
# separating by species
facet_wrap(~ ant_species,
           # labeling with sci_name
           labeller = labeller(ant_species = c("AB" = "C. sjostedti",
                                                "RRB" = "C. mimosae",
                                                "BBR" = "C. nigriceps")))) +

# setting colors by species
scale_color_manual(values = sp_cols) +
scale_fill_manual(values = sp_cols) +
# relabeling x and y-axes
labs(
  x = "Tree height (cm)",
  y = "Probability of nest occurrence"
) +
# changing theme
theme_bw() +
# removing legend
theme(legend.position = "none",
      # removing gridlines

```

```
panel.grid = element_blank()
```



i.

Figure 1. Model predictions for bird nest occurrence in acacia trees based on height and ant species Figure displays model predictions for probability of nest occurrence as tree height increases, separated by ant species living on the tree. Line represents exact predictions of model that assumes interaction between species and tree height. Ribbon represents model 95% confidence interval. Species are sorted by color with *C. sjostedti* being blue, *C. mimosae* being tan, and *C. nigriceps* being orange. Data originally from Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>.

j.

```
# creating prediction at 600 cm
ggpredict(model2,
  terms = c("height_cm [600]", "ant_species"))
```



```
# Predicted probabilities of case_control
```

```
ant_species: AB
```

height_cm	Predicted	95% CI
600	0.20	0.00, 0.99

```
ant_species: RRB
```

height_cm	Predicted	95% CI
600	0.74	0.48, 0.90

```
ant_species: BBR
```

height_cm	Predicted	95% CI
600	1.00	1.00, 1.00

```
# showing summary for baseline  
summary(model2)
```

```
Call:
```

```
glm(formula = case_control ~ height_cm * ant_species, family = "binomial",  
     data = nests_clean)
```

```
Coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.693410	2.517395	-1.467	0.14233
height_cm	0.003895	0.008237	0.473	0.63634
ant_speciesRRB	0.848334	2.554617	0.332	0.73983
ant_speciesBBR	-12.279738	5.946723	-2.065	0.03893 *
height_cm:ant_speciesRRB	0.002597	0.008386	0.310	0.75679
height_cm:ant_speciesBBR	0.084541	0.031961	2.645	0.00817 **

```
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)
```

```
Null deviance: 286.04  on 269  degrees of freedom
```

Residual deviance: 225.80 on 264 degrees of freedom
AIC: 237.8

Number of Fisher Scoring iterations: 7

k.

Ant species and tree height interact to predict the probability of a bird nest in a sample acacia tree. As tree height increases, nest probability increases. Additionally, the effect of height on nest occurrence probability is stronger for *C. nigriceps* ant species trees (odds ratio: 1.092) than *C. mimosae* ant species trees (odds ratio: 1.007) or *C. sjostedti* ant species trees (odds ratio: 1.004). At 600 cm (the highest recorded tree height), nest occurrence probability is 0.74 (95% CI: [0.48, 0.90]) for *C. mimosae*, 1.00 (95% CI:[1.00, 1.00]) for *C. nigriceps*, and 0.20 (95% CI: [0.00, 0.99]) for *C. sjostedti*.

It is important to note that the model predicted no significant interaction between tree height and *C. sjostedti*, (which accounts for the strange-looking figure and non-precise CI) meaning that the interaction between that specific species and tree height does not predict nest occurrence. This makes biological sense as *C. sjostedti* does not display defensive characteristics and therefore would not affect whether or not a nest is protected. *C. mimosae* acts as a good reference here because it is a good defender but does not alter the tree, meaning that there is interaction but it is more straightforward; especially when compared to *C. nigriceps* which is an aggressive defender but also alters tree structure to produce denser canopies therefore better defending nests (explaining sharp probability jump to 1 as height increases).

Problem 3. Problem 3. Affective and exploratory visualizations

a.

- My affective visualization is pretty much a completely new way to display data when compared to my initial exploratory plots. In effect, it is still a sort of scatter plot but point shape and color matter far more than location. There is also no models or metrics of central tendency as my goal with the affective visualization was not to make something that explained the whole dataset.
- Honestly it is pretty much impossible to see any similarities between my plots and my affective piece, they share no discernible characteristics besides the underlying data. I did separate by location in each case but I did so differently each time.
- Between the two exploratory plots there are some trends, mainly that walk time doesn't change too much on average between locations. There is also a pretty clear negative relationship between time left and total walk time that can be seen without any models.

- The main feedback I got was on improving readability. My visual was hard to understand but with time and explanation it became more legible for my peers. The visual improvements and readability improvements I made, upping line thickness, marking stops, using one line, etc., were all based on this conversation with my peers.

b.

Using walk_minutes as a response and week as the predictor I believe the most interesting analysis we can run is a t-test (or Mann-Whitney U if nonparametric) to see if mean (or median) walk time differs across location.

This is important to my central question of what is effecting my walk time to class, seeing if there is a difference across location could potentially invalidate my previous work on this data frame where I assumed non-differing central tendency between the two locations allowing for both locations' data to be used together.

Walk time is a continuous numeric variable.

Location is a categorical variable with 2 levels: Campbell (n = 19) and 387 (n = 13)

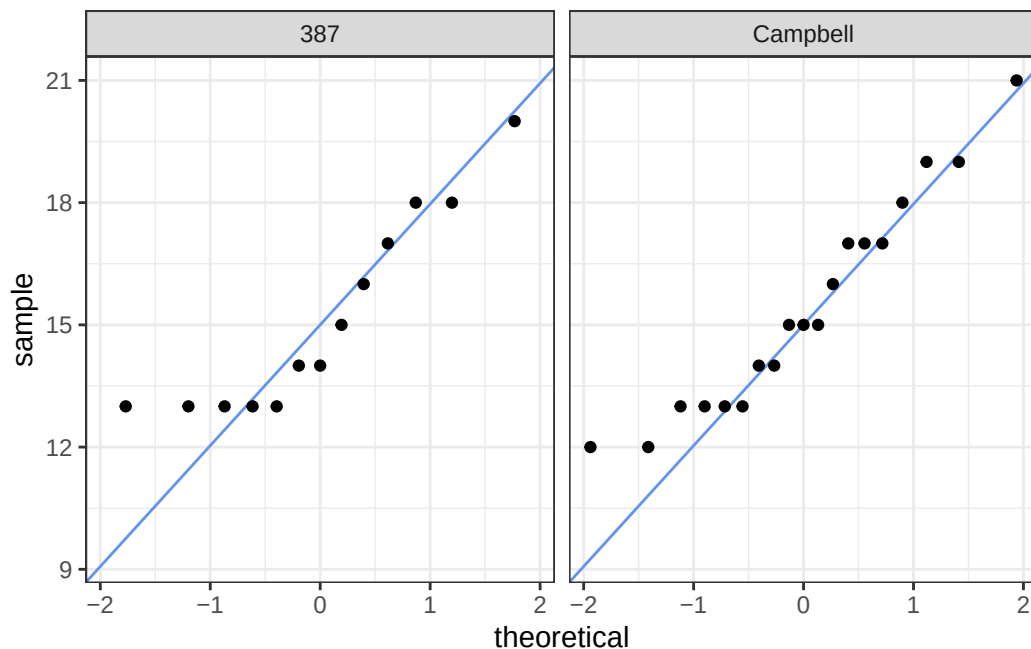
c.

Checking assumptions

- Observations are independent
- Response variable is continuous

Running QQ to test normality

```
# baselayer: ggplot
ggplot(data = personal_data, # using urchins clean data frame
       aes(sample = walk_minutes)) + # y-axis for QQ plot
# first layer: QQ reference line
geom_qq_line(color = "cornflowerblue") + # displaying line in blue
# second layer: QQ plot
geom_qq() +
# creating facets
facet_wrap(~location) +
# changing theme
theme_bw()
```



Not very normal looking, some grouping along line but likely need to use non-parametric due to low sample size.

Running analysis

```
# Mann-Whitney U test
# Walk_minutes response
# Location predictor
wilcox_test(walk_minutes ~ location,
             data = personal_data) # pulling from personal data
```

```
# A tibble: 1 x 7
  .y.      group1 group2    n1    n2 statistic     p
* <chr>    <chr>  <chr>   <int> <int>   <dbl> <dbl>
1 walk_minutes 387    Campbell    13    19     118.  0.83
```

Interpretation: Mann-Whitney test, $U = 118$, $p = 0.83$, $\alpha = 0.05$

No significant difference in rank distributions.

d.

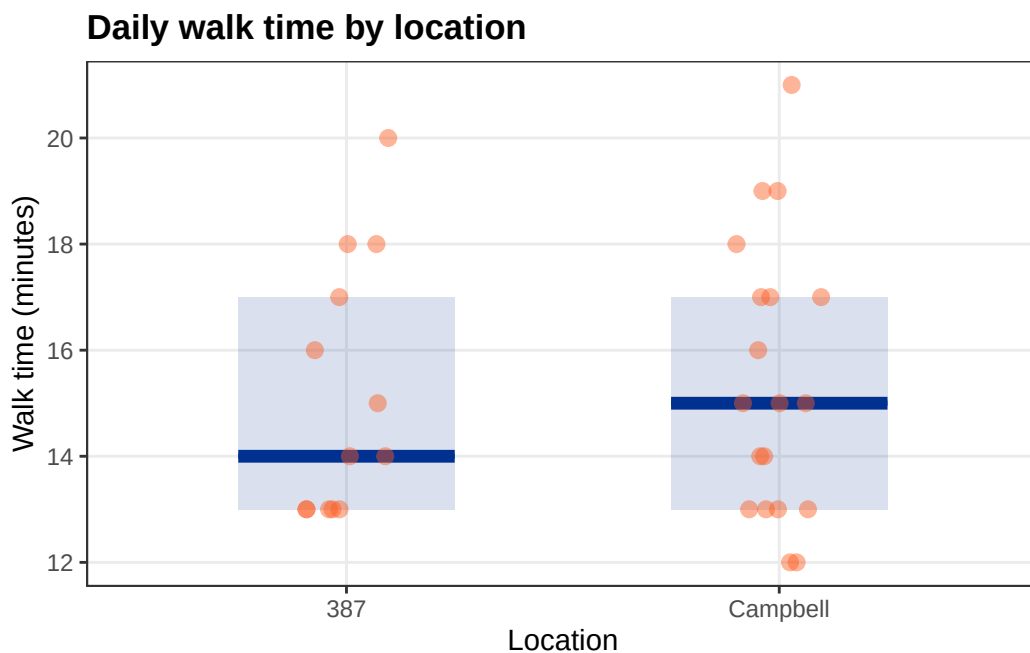
```
# creating new location median data frame
location_medians <- personal_data |>
  # grouping by location
  group_by(location) |>
  # summarizing data
  summarise(median_walk = median(walk_minutes), # calculated medians
            q25 = quantile(walk_minutes, 0.25), # calculated CI lower
            q75 = quantile(walk_minutes, 0.75)) |> # calculating CI Upper
  # ungrouping
  ungroup()

# base layer: ggplot
ggplot(data = personal_data,
       aes(x = location, # x-axis: location
           y = walk_minutes)) + # y-axis: walk minutes
  # first layer: rectangles for distributions
  geom_rect(data = location_medians,
            # setting rectangle's x max and min
            aes(xmin = as.numeric(factor(location)) - 0.25,
                xmax = as.numeric(factor(location)) + 0.25,
                ymin = q25, # ymin at lower quartile
                ymax = q75, # ymax at upper quartile
                x = NULL,
                y = NULL),
            # filling color
            fill = "#003090",
            # making slightly transparent
            alpha = 0.15) +
  # second layer: cross bars for medians
  geom_crossbar(data = location_medians,
               # setting to display medians for each location
               aes(y = median_walk,
                   ymin = median_walk,
                   ymax = median_walk),
               # coloring same as rectangle
               color = "#003090",
               width = 0.5, # make bar less wide
               linewidth = 0.9) + # making bar less thick
  # third layer: jitter for underlying data
  geom_jitter(height = 0, # removing vertical jitter
```

```

width = 0.1, # slight horizontal jitter
size = 2.5, # making points larger
alpha = 0.45, # making points transparent
color = "#FE5A1D") + # changing color
# relabeling title and axes
labs(title = "Daily walk time by location",
      x = "Location",
      y = "Walk time (minutes)") +
# changing theme
theme_bw() +
# further theme adjustments
theme(
  plot.title = element_text(face = "bold"), # bolding title
  panel.grid.minor = element_blank()) # removing minor gridlines

```



e.

Figure 2. No significant difference in walk time distributions across locations. Blue cross bars represent median walk time to class for each location (Campbell $n = 19$ and 387 $n = 13$) with the rectangle representing the IQR for each week's observations. Underlying data is shown via the orange points which each represent a single day's walk time. Data collected by Luke Simon between the dates of 2026-04-20 and 2026-06-03.

f.

From the dates of 2026-04-20 and 2026-06-03 there is no difference in daily walk time distributions between building 387 (median = 14, $n = 13$) and Campbell Hall (median = 15, $n = 19$) (Mann-Whitney test, $U = 118$, $p = 0.83$, $\alpha = 0.05$). Since our result is not significant we can assume that distributions are nearly identical (as is represented in the above figure) meaning that all previous analysis done and linear models created that do not differentiate between location are still valid. Essentially, walk times to each location are comparable and can be treated as independent observations in the same data set, any differentiation between the two locations would be included for clarity and aesthetic purposes only.